

Campus HelpDesk

AI-Powered Campus Support & Ticket Management Platform

Campus HelpDesk is a no-code AI-powered campus support platform designed to streamline communication between **students, staff, and administrators**.

The application enables students to raise support requests, automatically categorizes requests using AI, routes them to the appropriate department, tracks resolution progress, and collects feedback. Staff members can manage tickets using AI-assisted prioritization and summaries, while administrators can monitor campus-wide support performance through analytics and AI-powered feedback insights.

 **Live Demo:**

 **Live Application:**

<https://campus-solve-swift.base44.app>

 **GitHub Repository:**

<https://github.com/tripathimanu943-tech>

What Can Campus HelpDesk Do?

Campus HelpDesk provides three role-based experiences:

Student

Students can:

- Raise campus support requests
- Select or receive AI-suggested categories
- Submit requests and receive a unique ticket ID
- Track ticket progress through a visual timeline
- Communicate through ticket conversations
- Upload attachments
- Confirm resolution

- Provide star-based feedback
- Receive notifications about ticket updates

Staff

Staff members can:

- View requests assigned to their department
- Monitor New, In Progress, Waiting, Resolved and Overdue tickets
- Use AI to summarize requests
- Use AI to identify which requests should be handled first
- Monitor SLA countdowns
- Update ticket status
- Override AI recommendations when required
- Maintain an audit trail of staff overrides

Admin

Administrators can:

- Monitor overall campus support performance
- View KPI dashboards
- Analyze ticket volumes
- View department/category-wise analytics
- Search tickets using natural language
- Analyze student feedback using AI
- Identify recurring support issues
- Review SLA breaches
- Monitor ticket resolution performance

AI Features

The application demonstrates several practical applications of AI in campus management.

1. AI Auto-Categorization

When a student describes an issue, AI can suggest the appropriate category or department.

Example:

"The projector in Room 204 isn't working."

AI suggestion:

Department: IT

Category: Classroom Technology

The student can review and modify the AI suggestion before submitting the request.

2. AI Ticket Summarization

Staff can use AI to quickly summarize requests instead of reading the entire conversation.

Example:

Issue:

Projector malfunction in Room 204.

Impact:

Classroom presentation affected.

Recommended Action:

Assign to IT support and inspect the projector.

3. AI Priority / SLA Triage

AI assists staff in identifying requests that may require immediate attention.

Tickets can be flagged as:

- ☒ High Priority
- ☒ Medium Priority
- ☐ Low Priority

The system also displays SLA countdowns and overdue requests.

4. AI Feedback Analysis

Administrators can use AI to analyze student feedback and identify:

- Common complaints
 - Frequently mentioned issues
 - Positive experiences
 - Areas requiring improvement
 - Recurring department-level problems
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5. Natural-Language Search

Administrators can search campus support data using natural-language questions.

Example:

"Show me all overdue IT requests."

The system can interpret the query and surface relevant tickets.

Ticket Lifecycle

The core ticket workflow is:

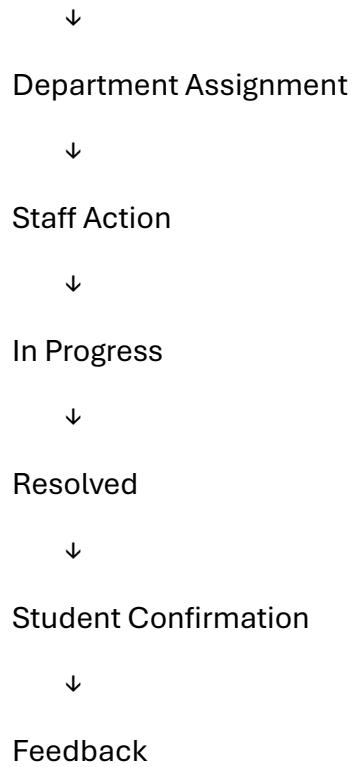
Student Raises Request



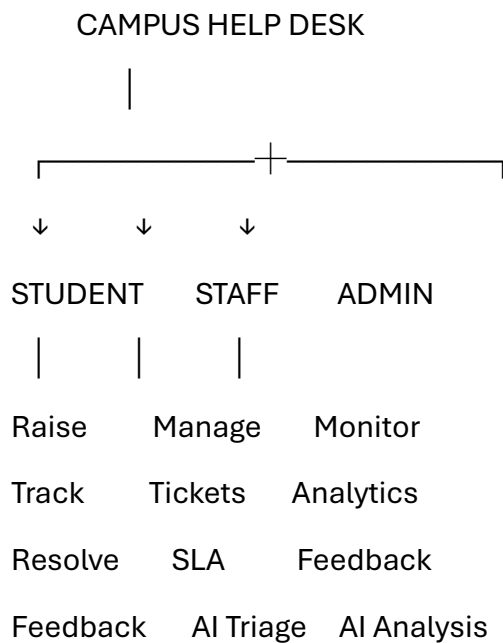
AI Categorization



Ticket Created



Role-Based Architecture



The application includes a **demo role switcher** that allows users to experience the Student, Staff, and Admin interfaces.

User Experience

The application was designed with a modern SaaS-style interface.

Design Features

- Indigo/violet gradient CTAs
- Department-based color coding
- Responsive design
- Mobile bottom navigation
- Dark mode support
- Visual ticket timelines
- Status badges
- SLA countdown indicators
- Notification system
- Interactive dashboards
- Responsive role-based interfaces

Demo Dataset

The application includes **32 seeded demo tickets** distributed across **9 departments**.

The demo dataset includes:

- Active tickets
- Resolved tickets
- Overdue tickets
- Comments
- Student feedback
- SLA breaches
- Historical submission timestamps

This allows the dashboards and analytics to demonstrate realistic campus-support scenarios without requiring real student data.

 Possible Campus Use Cases

Campus HelpDesk can be adapted for multiple university support functions:

Department	Example Request
IT	Wi-Fi or projector issue
Library	Book/resource issue
Administration	Document request
Finance	Fee-related query
Hostel	Maintenance request
Transport	Bus-related issue
Academics	Academic support
Facilities	Classroom maintenance
Student Services	General student support

 Technology & Development Approach

Component	Technology
Application Platform	Base44
Development Approach	No-Code
AI	AI-assisted workflows
Application Type	Campus Support Platform
User Roles	Student / Staff / Admin

Component	Technology
Data	Demo/Seeded Dataset
Interface	Responsive Web Application

Note: Campus HelpDesk was developed as a no-code application using Base44. The project demonstrates how AI and no-code tools can be combined to build practical solutions for education and campus operations.

Project Structure

campus-helpdesk/

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├─ README.md

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├─ screenshots/

|

│ └─ student-dashboard.png

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│ └─ raise-request.png

|

│ └─ ticket-tracking.png

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│ └─ staff-dashboard.png

|

└─ admin-dashboard.png

|

├─ documentation/

|

│ └─ project-overview.md

|

│ └─ workflow.md

|

└─ ai-features.md

|

└─ assets/

└─ campus-helpdesk-banner.png

Project Objective

The objective of Campus HelpDesk is to demonstrate how **Artificial Intelligence, no-code application development, and workflow automation** can be applied to solve real-world problems in higher education.

Rather than functioning only as a chatbot, the platform demonstrates an end-to-end support workflow:

AI → Categorization → Routing → SLA Management → Resolution → Feedback → Analytics

Key Learning Outcomes

This project demonstrates practical experience in:

- Applied AI
 - Generative AI
 - AI-assisted decision support
 - No-code application development
 - Workflow design
 - AI-based categorization
 - AI summarization
 - Natural-language interfaces
 - Dashboard design
 - Education technology
 - Student support automation
 - SLA-based ticket management
 - AI feedback analysis
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Project Creator

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★ Project Highlights

Campus HelpDesk is a practical demonstration of how no-code development and AI can be combined to build an intelligent student-support ecosystem for higher education institutions.

Project Type

Applied AI + EdTech + No-Code

Primary Users

Students | Staff | Administrators

Core Concept

Raise → Categorize → Route → Track → Resolve → Analyze

Campus HelpDesk — App Build Prompt (for Base44)

Build a functional web app called ****Campus HelpDesk**** — a centralized platform where students raise campus problems (ID card, Wi-Fi, classroom, electrical, hostel, library, transport, exams, finance, etc.), AI auto-categorizes and routes each request to the right department, and everyone tracks it through to resolution.

Core loop: ****Raise → Route → Track → Resolve → Feedback****

1. Design Theme

Modern, clean SaaS product — not a plain complaint form.

- **Palette:** Deep indigo/navy primary (#1E3A8A or similar) + white/light-gray backgrounds + one accent color (teal or amber) for CTAs. Status colors: yellow = open, blue = in progress, green = resolved, red = urgent/emergency.
- **Style:** Rounded cards (12–16px radius), soft shadows, generous white space, clear typography hierarchy (bold headings, readable body text).
- **Components:** Pill-shaped status badges, progress timelines with connected dots, summary stat cards with icons, sticky/prominent primary button.
- **Responsive:** Mobile-first — most students will use this on their phones.
- The single most important element on screen is the **"+ Raise a Request"** button — make it impossible to miss.

2. User Roles & Access

Role-based login (Student / Staff / Admin) — dashboard shown automatically depends on role.

| Role | Can do |

|---|---|

| ****Student**** | Create requests, view/track own requests only, comment, upload attachments, confirm resolution, reopen, give feedback |

| ****Staff**** | View/manage requests assigned to their department only, update status, comment, mark resolved, see dept. performance |

| ****Admin**** | View everything, manage users/departments/categories, reassign requests, override priority, view analytics |

3. Data Model

****Users**** — id, name, email, role, department, course, semester

****Departments**** — id, name, description, contact

****Categories**** — id, name, linked department (used for auto-routing)

****Tickets**** — id, student_id, title, description, category, department, location, priority, status, assigned_staff, ai_category, ai_department, ai_priority, ai_summary, created_at, updated_at, resolved_at, closed_at

****Comments**** — id, ticket_id, user_id, comment, created_at

****Notifications**** — id, user_id, ticket_id, message, read_status, created_at

****Feedback**** — id, ticket_id, student_id, rating (1–5), comment, created_at

Ticket ID format: ****CHD-1001****, incrementing.

4. Student Flow

1. **Dashboard:** "How can we help you today?" header → big **+ Raise a Request** button → 4 stat cards (Open / In Progress / Resolved / Urgent) → list of recent tickets (ID, title, department, status badge, priority, date, "Track Request" link).
2. **Raise a Request form:** Title, Description, Category (dropdown, 15 options incl. "Other"), Location (dropdown + custom text), Priority (Low/Medium/High/Emergency, with an AI-suggested default), optional file/image upload.
3. **AI Analysis:** On submit, AI reads the description and returns: category, department, priority, a one-line issue summary, and a suggested action for staff. Show this to the student as a quick confirmation step before final submission ("Here's what we found — confirm or edit").
4. **Confirmation screen:** "Request Submitted 🎉" with ticket number, department, priority, expected response time, and a "Track My Request" button.
5. **Tracking page:** Vertical timeline — Submitted → Assigned → Under Review → In Progress → (Waiting for Student) → Resolved → Closed. Current step visually highlighted.
6. **Ticket conversation:** Simple chat-style thread between student and staff.
7. **Resolution confirmation:** When staff marks resolved, prompt student: "Is your issue resolved?" → Yes (close) / No (reopen + ask what's still wrong).
8. **Feedback:** After closing, star rating (1–5) + optional comment.

5. Staff Flow

- Dashboard scoped to their department: New / In Progress / Waiting for Student / Resolved Today counts, plus a request list (ticket, title, priority) with **View / Assign / Update Status** actions.
- Can see and override AI-suggested category/department/priority.
- **"💡 Summarize My Requests"** — AI generates a short daily briefing (open count, high-priority count, requests stale >24h, most common issue today, which tickets need immediate attention).

- **"What Should I Handle First?"** — AI ranks open tickets by priority, age, and impact.

6. Admin Dashboard

- KPI cards: Total Requests, Open, In Progress, Resolved, Avg. Resolution Time, Satisfaction Score.

- Charts: requests by category, by department, by status, over time, avg. resolution time, satisfaction by department.

- Searchable/filterable/sortable ticket table: Ticket ID, Student, Category, Department, Priority, Status, Created, Assigned Staff, Resolution Time, Actions. Filters for department, category, priority, status, date range.

- Manage: departments, categories, category→department routing rules, users.

- **AI Feedback Analyzer:** overall satisfaction %, top recurring complaints, per-department rating, one AI-generated operational recommendation (e.g., "Facilities has the highest avg. resolution time — consider added staffing at peak hours").

7. Automatic Routing Rules (editable by admin)

| Category | Department |

|---|---|

| ID Card | Administration |

| Wi-Fi / IT | IT Department |

| Classroom / AC | Facilities |

| Library | Library |

Transport	Transport Department
Finance / Fees	Finance
Examination	Examination Cell
Hostel	Hostel Administration
Academic	Academic Department

8. AI Guardrails

- Always label AI output as "AI Suggested" — never auto-final.
- Staff/admin can edit AI category, department, and priority at any time.
- AI must only use real ticket/data on record — never fabricate ticket details. If data is missing, show "No data available" rather than guessing.
- Add a visible note: *"AI suggestions are recommendations and should be verified by authorized staff."*
- Natural-language search bar ("show unresolved Wi-Fi complaints", "which requests are open >2 days") — AI interprets and filters real ticket data only.

9. Notifications

In-app notifications for: request submitted, assigned, status change, staff comment added, more info requested, resolved.

10. Demo Data (seed on build)

- 20 students, 10 staff across 9 departments (IT, Administration, Facilities, Library, Transport, Finance, Examination, Academic Affairs, Hostel).
- 30+ sample tickets spanning all categories, priorities, statuses, and a spread of dates so dashboards look populated immediately.

11. Priorities

Build in this order: (1) Student request creation + AI categorization, (2) ticket tracking timeline, (3) staff dashboard + status updates, (4) admin dashboard/analytics, (5) feedback loop, (6) AI staff summaries and natural-language search.

Keep every flow to the fewest possible clicks. Prioritize a working end-to-end demo (the projector/IT example) over extra features.